

JAVA DAY 2

PRACTICE QUESTIONS

60+ Questions – Arrays, Strings, Methods, Recursion, Class/Object, Constructors, this/static/final, Encapsulation, Inheritance

Attempt without notes first. Write code on paper and dry-run. Practice explaining theory answers out loud.

1. 1D Arrays – Traversal

Q1. How do you declare and initialize a 1D array in Java? Give three different ways.

Q2. What is the difference between `arr.length` and a method `length()`?

Q3. Write a program to find the maximum and minimum element in an array using a single traversal.

Q4. Predict the output:

```
int[] a = new int[3];
a[0] = 10; a[1] = 20;
System.out.println(a[2]);
System.out.println(a.length);
```

2. Array Operations – Insert, Delete, Search

Q5. Write a linear search method that returns the index of a key or -1 if not found.

Q6. How do you insert an element at a given index in a fixed-size array (using a logical size)?

Q7. Write code to delete the first occurrence of a value from an array (shift elements).

Q8. What is the time complexity of linear search vs binary search? When can binary search be used?

3. 2D Arrays – Matrix Traversal

Q9. How is a 2D array stored in memory in Java? What does `mat.length` and `mat[0].length` give?

Q10. Write nested loops to print a matrix in row-major order.

Q11. Write a program to calculate the sum of each row and each column of a matrix.

4. Matrix Operations – Transpose

Q12. Define matrix transpose. What is the size of the transpose of an $m \times n$ matrix?

Q13. Write a program to find the transpose of a rectangular matrix.

Q14. How can you transpose a square matrix in-place? Write the code.

5. String Basics – Immutable Nature

Q15. Why are Strings immutable in Java? Give at least three reasons.

Q16. What is the String Pool? Explain the difference between `String s = "Hi"` and `String s = new String("Hi")`.

Q17. Predict the output:

```
String s1 = "Java";
String s2 = "Java";
String s3 = new String("Java");
System.out.println(s1 == s2);
System.out.println(s1 == s3);
System.out.println(s1.equals(s3));
```

6. String Methods – API Usage

Q18. List any 8 commonly used String methods with a one-line description each.

Q19. What is the difference between `==` and `equals()` for Strings?

Q20. Write code to reverse a string without using `StringBuilder/Buffer`.

Q21. How does `substring(begin, end)` behave? Is the end index inclusive?

7. StringBuilder – Mutable Strings

Q22. Why is `StringBuilder` preferred over `String` concatenation inside a loop?

Q23. Write a program using `StringBuilder` to reverse a string and then convert it back to `String`.

Q24. Name five important methods of `StringBuilder`.

8. StringBuffer – Thread Safety

- Q25. What is the main difference between StringBuilder and StringBuffer?
- Q26. When should you use StringBuffer instead of StringBuilder?
- Q27. What is the default capacity of StringBuffer/StringBuilder? How does capacity grow?

9. Advanced Strings – Pattern Matching

- Q28. Write a method to check whether a string is a palindrome.
- Q29. How can you count the frequency of each character in a string?
- Q30. Write a simple regex check (using matches) for a basic email pattern.

10. Methods & its types – Creation

- Q31. What is method overloading? Give an example with different parameter lists.
- Q32. Can two methods differ only by return type and still be overloaded? Why?
- Q33. Differentiate between instance method and static method.

11. Pass by Value – Parameter Passing

- Q34. Is Java pass-by-value or pass-by-reference? Explain with primitives and objects.
- Q35. Why does a swap method that only reassigns parameters fail to swap the caller's variables?
- Q36. Predict the output:

```
class Num { int val; Num(int v){val=v;} }
static void change(Num n){ n.val = 99; n = new Num(0); }
public static void main(String[] args) {
    Num x = new Num(5);
    change(x);
    System.out.println(x.val);
}
```

12. Recursion Intro – Call Stack

- Q37. What is a base case? What happens if a recursive method has no base case?
- Q38. Explain the call stack for factorial(4).
- Q39. Write a recursive method to calculate the sum of digits of a number.
- Q40. Write a recursive method to print numbers from N down to 1.

13. Class & Object

- Q41. What is the difference between a class and an object?
- Q42. What are the default values of instance variables of type int, boolean, and String?
- Q43. How many objects are created in: Student s1 = new Student(); Student s2 = s1;?

14. Constructors – Default, Parameterized

- Q44. What is a constructor? How is it different from a normal method?
- Q45. If you write a parameterized constructor, does the compiler still provide a default constructor?
- Q46. Write a class with a default constructor and two overloaded parameterized constructors.

15. this Keyword

- Q47. List three uses of the this keyword.
- Q48. What is constructor chaining? Write an example using this().
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Q49. Can `this()` and `super()` both appear in the same constructor? Why?

16. static Keyword

Q50. What is a static variable? How many copies exist for a class?

Q51. Can a static method access an instance variable directly? Explain.

Q52. When is a static block executed? Write a small demo.

17. final Keyword

Q53. What happens when a variable / method / class is declared `final`?

Q54. Can a final method be overloaded? Can it be overridden?

Q55. What is a blank final variable and where can it be initialized?

18. Encapsulation – Getters & Setters

Q56. Define encapsulation. How do getters and setters help achieve it?

Q57. Write a `BankAccount` class with private balance and proper deposit/withdraw methods (with validation).

Q58. Why might you provide only a getter and no setter for a field?

19–20. Inheritance (Single, Multilevel, Hierarchical)

Q59. What is inheritance? What keyword is used? Does Java support multiple inheritance of classes?

Q60. Differentiate between multilevel and hierarchical inheritance with a small diagram or description.

Q61. What is method overriding? What is the role of the `super` keyword?

Q62. Predict the output:

```
class A { void show(){ System.out.println("A"); } }
class B extends A { void show(){ System.out.println("B"); } }
public class Test {
    public static void main(String[] args) {
        A obj = new B();
        obj.show();
    }
}
```

Bonus / Mixed

Q63. Write a recursive binary search method on a sorted array.

Q64. Create a class `Employee` with encapsulation, a parameterized constructor, and a static counter of employees.

Q65. Explain why `String` is `final` and immutable. How does this help the String Pool?

After attempting, cross-check with the Day 2 Concept Guide PDF. Focus on dry-running arrays, recursion, and inheritance hierarchies.

All the best for your placements!